

Advanced (Habanero)

Q1. Solve the equation $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve the equation $|2x| = 8$.

- (A) $x = 2$ or $x = -2$
- (B) $x = 4$ or $x = -4$
- (C) $x = 1$ or $x = -1$
- (D) $x = 8$ or $x = -8$
- (E) $x = 0$

Q3. Solve the equation $|x - 2| = 3$.

- (A) $x = 5$ or $x = -1$
- (B) $x = 2$ or $x = -2$
- (C) $x = 1$ or $x = -1$
- (D) $x = 8$ or $x = -8$
- (E) $x = 0$

Q4. Solve the equation $|3x + 2| = 11$.

- (A) $x = 3$ or $x = -5$
- (B) $x = 1$ or $x = -1$
- (C) $x = 2$ or $x = -2$
- (D) $x = 9$ or $x = -9$
- (E) $x = 0$

Q5. Solve the equation $|x + 4| = 2$.

- (A) $x = -2$ or $x = -6$
- (B) $x = 2$ or $x = -2$
- (C) $x = 1$ or $x = -1$
- (D) $x = 8$ or $x = -8$
- (E) $x = 0$

Q6. Solve the equation $|2x - 5| = 9$.

- (A) $x = 7$ or $x = -2$
- (B) $x = 1$ or $x = -1$
- (C) $x = 2$ or $x = -2$
- (D) $x = 9$ or $x = -9$
- (E) $x = 0$

Q7. Solve the equation $|x^2 - 4| = 8$.

- (A) $x = 2$ or $x = -2$
- (B) $x = 4$ or $x = -4$
- (C) $x = 1$ or $x = -1$
- (D) $x = 6$ or $x = -6$
- (E) $x = 0$

Q8. Solve the equation $|3x + 1| = |2x - 5|$.

- (A) $x = 2$ or $x = -2$
- (B) $x = 1$ or $x = -1$
- (C) $x = 3$ or $x = -3$
- (D) $x = 4$ or $x = -4$
- (E) $x = 3$

Open Ended Questions (FRQ)

Q9. Solve the equation $|x + 2| + |x - 3| = 5$ and check for extraneous solutions.

Q10. Solve the equation $|2x + 1| - |x - 2| = 3$ and check for extraneous solutions.

Chef's Tips

- Tip 1: Remind students to check for extraneous solutions.
- Tip 2: Suggest that students use the properties of absolute value to simplify the equations.
- Tip 3: Encourage students to use algebraic methods to solve the equations.